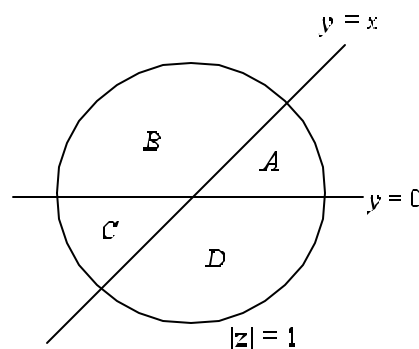


Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!**

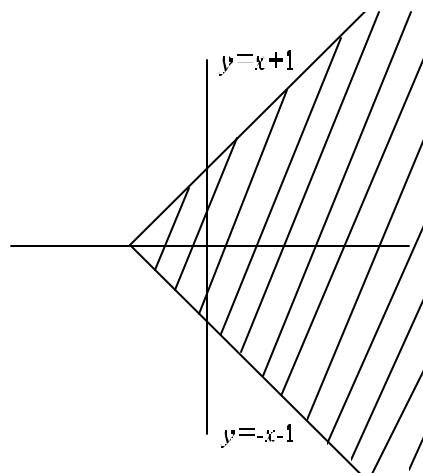
Notation: $\mathbb{C}$ = set of complex numbers
 $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$
 $\text{RHP} = \{z \in \mathbb{C} : \text{Re } z > 0\}$
 $A(\mathbb{D}) = \{f : f \text{ is analytic on } \mathbb{D}\}$

1. Find the bilinear transformation M which maps 1, -1, 0 to 0, 1, ∞ resp. Describe the images of A, B, C, D under M where A, B, C, D are given in the diagram to the right.

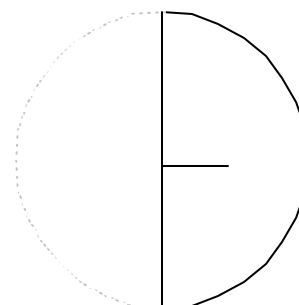


2. Suppose that f is entire and that $\lim_{z \rightarrow \infty} f(z)$ exists. Show that f is constant.

3. Let G be the open sector described in the diagram to the right. Find a one-to-one, onto conformal map f which maps G to $\mathbb{D}$ such that $f(0) = 0$.



4. Let G be the region $\{\mathbb{D} \cap \text{RHP}\} \setminus (0, \frac{1}{2}]$ (see figure to right). Find a one-to-one, onto conformal map f which maps G to $\mathbb{D}$ such that $f(\frac{3}{4}) = 0$.



5. Evaluate the integral $\int_g (z + \bar{z}) dz$ where $g(t) = e^{it}$, $0 \leq t \leq p$
6. Evaluate the integral $\int_g \frac{dz}{z}$ where $g(t) = e^{it}$, $0 \leq t \leq \frac{3p}{2}$
7. Evaluate the integral $\int_g \frac{e^{-z} - 1}{(1+z)^2} dz$ where $g(t) = 2e^{it}$, $0 \leq t \leq 2p$
8. Evaluate the integral $\int_g \frac{z^2 + 4}{z^2 - 1} dz$ where $g(t) = 1 + e^{it}$, $0 \leq t \leq 2p$
9. Let $G = \mathbb{C} \setminus \{z = k\frac{p}{2} : k \text{ odd integer}\}$. Prove the identity that for all $z \in G$ that $1 + \tan^2 z = \sec^2 z$. [Do not prove this using the identity that for all $z \in \mathbb{C}$ that $\sin^2 z + \cos^2 z = 1$.]
10. Let $f \in A(\mathbb{D})$ such that $f : \mathbb{D} \rightarrow \mathbb{D}$. Show that for any positive integer n that $|f^{(n)}(0)| \leq n!$
11. State and prove the Fundamental Theorem of Algebra.